

9708/31/ may june 2022

1-C:

Pareto optimality is named after **Vilfredo Pareto**, an Italian economist who first introduced the concept. It's a state where resources are allocated in such a way that:

1. No individual can be made better off without making someone else worse off.
2. There is no possible reallocation of resources that could improve someone's situation without deteriorating someone else's.

To **sum up**, **Pareto optimality** is a state where resources are allocated in such a way that no individual can be made better off without making someone else worse off. This means that any reallocation of resources would lead to a loss for at least one person, making it impossible to improve someone's situation without deteriorating someone else's.

Conclusively, this represents that we are operating on the boundary of the production possibility curve. This is why, key C, becomes correct.

A) Consumers maximizing their utility represents "consumer equilibrium"

B) This statement describes the concept of "productive efficiency"

D) Pareto efficiency does not mean that potential losers from any re-allocation of resources cannot be compensated by those who gain.

2-C:

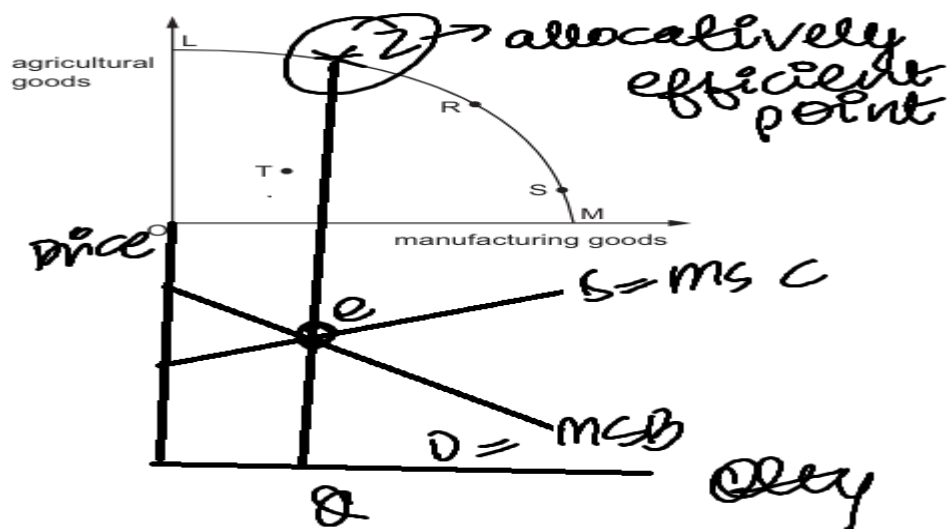
The **production possibility curve** (PPC) shows the various combinations of agricultural goods and manufacturing goods that the economy can produce given its resources. **Point S is on the PPC, indicating that the economy is fully utilizing its resources to produce a combination of both goods.** This means that the economy is productively efficient, as it is producing on the frontier of its possibilities.

Note: All points on the PPC are productively efficient.

This explains why key C is correct.

Here's why the other options are incorrect:

- A) Point R is not necessarily allocatively efficient, as it may not represent the optimal combination of goods based on consumer preferences. **This is because, while all points on the PPC are productively efficient, only one point is allocatively efficient. The point which is productively efficient might not be allocatively efficient but the point which is allocatively efficient is necessarily productively efficient.**
- > **This can be better understood through the diagram below, which suggests that point R may not be allocatively efficient.**



B) Since point T is inside the PPC, this does not represent a pareto efficient point as it's possible to reallocate resources to make someone better off without making someone else worse off.

D) The opportunity cost is not constant as we move from L to M. The PPC is concave or bowed outwards, indicating that the opportunity cost of producing more of one good increases as we move along the curve. **(Increasing opportunity cost)**

3-C:

Since the private benefit is greater than the private cost, the activity is profitable for the individual or firm undertaking it. However, the social benefit ($pb+eb$) is less than the social cost ($pc+ec$), indicating that the activity has negative externalities (external costs) that outweigh any external benefits.

This means that the external costs exceed the external benefits, making option C the correct answer.

4-C:

Consumers are in equilibrium when the ratio of the **marginal utility of goods X and Y are equal to the ratio of their prices.**

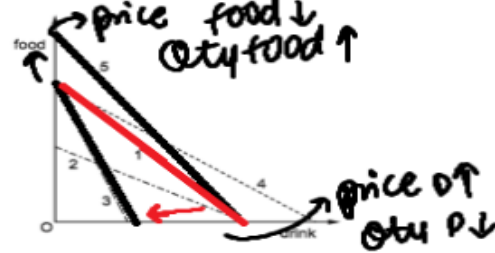
The central principle of marginal utility theory, also known as the "equi-marginal principle," states that a consumer is in equilibrium when the ratio of the marginal utility per dollar spent on each good is equalized i.e. $MUX/PX = MUY/PY$. This means that the consumer is getting the same additional satisfaction (marginal utility) from the last unit of each good, relative to their prices.

In other words, the consumer is allocating their budget in such a way that the marginal utility per dollar spent is equal for all goods.

This is why, only key C, can be correct.

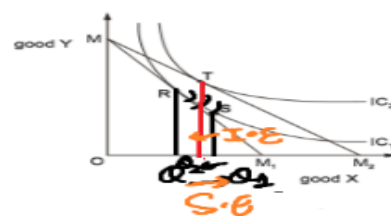
5-C:

The diagram shows five budget lines. Line 1 is the original budget line.



- In case of a move to budget line "3", the price of drinks is shown to increase, which is why, there is a contraction in qty demanded of drinks.
- Similarly, in case of a move to the budget line "5", the price of food is shown to decrease, which is why, there is an extension in the qty demanded of food. Since food has now become relatively cheaper, drink will have a "relatively higher price", or in other words, it will appear relatively expensive.
- This is why, key C, becomes correct.

6-B:



* Initial Budget line = MM_1
 * price x falls :
 Budget line tilts rightwards to MM_2

* Income effect :
 movement from S to T .
 contraction in qty demanded from Q_1 to Q_2

* Substitution effect :
 extension in qty demanded from Q_1 to Q_2 .
 (movement along the indifference curve from R to S)

Because of a fall in the price of inferior good, according to the substitution effect, the qty demanded extends or rises.

But since, a fall in price brings about a rise in real income, it results in a contraction or a fall in the qty demanded of the inferior good, according to the income effect

7-D:

- A) Similar cost structures make it easier for firms to agree on a common price.
- B) Close substitutes mean that firms are more interdependent and have a stronger incentive to coordinate prices.
- C) High barriers to entry make it easier for incumbent firms to maintain their market power and coordinate prices without worrying about new entrants disrupting the market.
- D) Collusion requires coordination and agreement among firms, which is more difficult to achieve when there are many firms in the market. In a market with a large number of competing firms, it is more likely that one firm will deviate from the agreed-upon price, making the agreement weak. This explains why key D will be correct.

8-C:

Ambitious risk-takers are more likely to pursue growth and expansion opportunities, rather than choosing to remain small. The other options are more plausible reasons why firms might choose to remain small despite potential economies of scale:

- A) Frequent and rapid changes in demand make it difficult to scale up production and adjust to new market conditions.
- B) Entrepreneurs may prefer to maintain personal control and avoid the complexities and risks that come with growth.
- D) Specialized markets with unique products may have limited demand, making growth and scaling up less viable or necessary.

9-C:

C) In the long-run equilibrium of a monopolistically competitive market, firms earn zero economic profit, meaning that the profit-maximizing price (AR) equals average cost (AR=AC).

The other options are likely to occur in the long-run equilibrium:

- A) The individual firm's demand curve becomes more elastic as more firms enter the market in the long-run after having been attracted to short-run supernormal profits, increasing competition.
- B) The individual demand curve faced by a typical firm may shift left as new firms enter the market after having been attracted to short-run supernormal profits.
- D) The profit-maximizing price is indeed greater than marginal cost ($P > MC$) in the short-run equilibrium, and this condition continues to hold in the long-run equilibrium. This is because a monopolistically competitive firm is not allocatively efficient (price is not equal to MC)

10-B:

Key Takeaway : Government provision is often considered when the market fails to provide essential goods or services efficiently.

B - Direct provision of a product by the government is least likely when the market can efficiently supply the product at a low cost. High demand and low unit costs indicate a competitive market with many suppliers, making government intervention unnecessary.

- A) High fixed costs could lead to a natural monopoly, making government provision more feasible.
- C) Large natural barriers to entry could limit competition , justifying government intervention to prevent monopolistic exploitation.
- D) Minimum efficient scale of production above current demand suggests that the private firms might not produce a certain product because they would not be able to achieve productive efficiency given the current level of demand. This implies that government intervention will be necessary in this scenario.

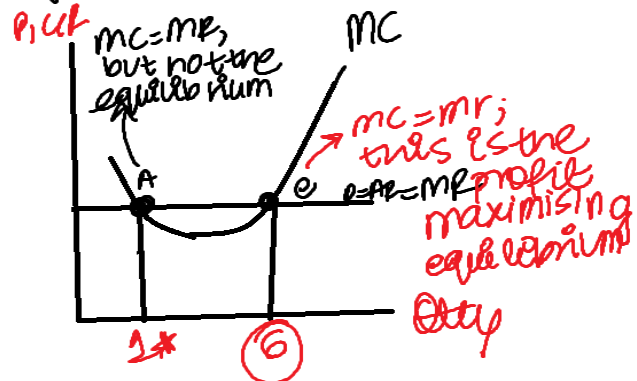
11- C:

Key takeaway: In case of perfect competition, **market price (AR) is equal to Marginal revenue, because price remains the same.** This explains that the Marginal revenue will be "8"

Units	Total Cost	Marginal Cost	Marginal Revenue	Profit Max?
1	8	8	8	No
2	14	6	8	No
3	18	4	8	No
4	22	4	8	No
5	28	6	8	No
6	36	8	8	Yes
7	46	10	8	No
8	58	12	8	No

$MC = MR$

see the diagram



12-C:

- A) This option is incorrect because increasing government regulation can be costly and burdensome for small firms, making it harder for them to operate and grow.
- B) This option is also incorrect because more risk-averse lending policies reduce access to credit for small firms, making it harder for them to secure funding and expand their business.
- C) This option is correct because stronger antitrust laws and enforcement against monopolies can lead to a more competitive market, allowing small firms to thrive. This can result in the breakup of large monopolies, increased competition, and encouragement of new entrants, including small firms.
- D) This option is incorrect because tougher penalties for personal bankruptcy make entrepreneurship riskier, discouraging new ventures.

13-B:

A: Imperfect information is a source of market failure. -> If individuals are not aware of the benefits or positive externalities associated with consuming merit goods, they may not prioritize their consumption, even if it is in their best interest or contributes to the overall welfare of society. For example, preventive healthcare services or education may be underutilized if individuals are not fully informed about their long-term benefits. This under-allocation of merit goods results in market failure.

If individuals are unaware of the harmful consequences or negative externalities associated with demerit goods, they may over- consume them without fully considering the long-term repercussions. For example, individuals may engage in excessive consumption of harmful substances such as: alcohol, due to a lack of information about the health risks involved. This over -allocation of de-merit goods results in market failure.

B - Income inequality refers to the unequal distribution of income among individuals or households in an economy. This is a macro-economic concern and may not cause market failure. This is why, key B, becomes correct.

C) Monopoly is a source of market failure. A “monopoly” occurs when a single firm dominates the market and restricts competition. In a monopoly, the monopolistic firm has significant market power, allowing it to set prices higher than in a competitive market, resulting in reduced output and consumer welfare. This results in “market failure”.

D) Non-excludability is a source of market failure, particularly concerning public goods. Public goods are goods or services that are non-excludable, meaning that payment by one individual does not restrict others from enjoying the benefits associated with these goods; even the non-tax payers get to consume it. This results in the free-rider problem. Due to the free-rider problem, where individuals can benefit from public goods without paying for them, private markets do not provide for them, and this non-provision of public goods results in market failure.

14-B:

The impact on profits will be:

- Buyers of permits: Decrease in profits
- Sellers of permits: Increase in profits

Buyers of permits: Previously, they had a free permit, but now they have to buy one from another firm. This added cost reduces their profits.

Sellers of permits: They can now sell their excess permits, generating incomes and increasing their profits.

By allowing permits to be traded, the government has created a market for pollution credits, which adds a new revenue stream for some firms (sellers) and a new cost for others (buyers).

15-D:

Nudge theory aims to influence behavior in predictable ways without limiting freedom of choice. In this case, placing fruit near cash registers is a subtle cue that nudges consumers towards making a healthier choice, increasing the likelihood of impulse purchases of fruit.

Key take-away: By strategically placing fruit near cash registers, supermarkets are using a gentle nudge to encourage healthier choices, making it more likely that consumers will grab some fruit along with their other purchases.

A) Advertising the benefits of healthy eating is more of an informational campaign than a nudge.

B) Increasing tax on sugary foods involves creating financial dis-incentives to discourage consumption of sugary drinks.

C) Introducing a subsidy for fruit growers is a policy change that creates financial incentives for producers to produce fruits.

16-B:

Transfer earnings refer to the minimum amount of earnings required to keep a factor of production (like labor) in its current use or job. It's the amount needed to prevent a worker from switching to a different job or industry.

A) This definition is actually describing economic rent, which is the amount earned above the transfer earnings.

C) Social security benefits are a type of government support, not related to transfer earnings.

D) The amount of earnings needed to cause a worker to change to a different job does not represent "transfer earnings"

17-D:

D) As workers become more productive, an additional labor hired generates higher value, which is why, the demand for labor or marginal revenue productivity increases. This results in an upward pressure on wages.

A) An increase in capital-intensive production might actually lead to lower labor demand and, consequently, lower wages.

B) An increase in the number of industrial workers would likely lead to a surplus of labor, putting downward pressure on wages.

C) An increase in the number of people unemployed would also put downward pressure on wages, as a larger labor supply relative to demand tends to drive wages down.

18-A:

The table shows that Brazil's economic growth rate was negative in 2015 (-3.5%), indicating a contraction in the economy, and since this has been there for two successive quarters, this represents a recession, which is typically considered a downturn in the business cycle.

This explains why key A will be correct.

Here's why the other options are incorrect:

B) Japan's real GDP was not at its smallest in 2014. While the growth rate was low (0.4%), the actual GDP would have been high because it has been growing every year until 2014, after which it started falling.

C) Spain did experience a recession in 2012 and 2013, but it was not the worst recession of the period. Brazil's recession in 2015 and 2016 was more severe.

D) Spain was not in the downturn of its trade cycle in 2014. In fact, Spain's growth rate was positive in 2014 (1.4%), indicating an expansion.

19-D:

Official statistics for national income per head may **overstate** the true level of economic well-being in a developing country if there is extreme income inequality. This is because the average income per head does not reflect the distribution of income among the population. If a small percentage of upper income groups hold a significant portion of the wealth, the average income per head may be skewed, hiding the fact that many people live in poverty.

This explains why key D is correct.

A) Barter in internal trade would likely lead to underreporting of national income, as these transactions may not be captured in official statistics.

B) Subsistence agriculture dominating economic activity would be likely to understate the level of national income per head.

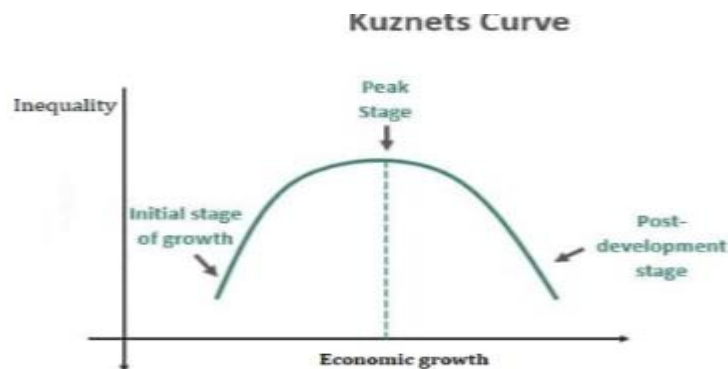
C) Exports in services are included in the national income per head data.

20-C:

A - The J-curve effect suggests that a depreciation of the currency will initially lead to a worsening trade balance (higher deficit) before eventually improving it.

B) The economic growth rate may rise and fall over time, but this is not the specific relationship shown by the Kuznets curve.

C) The Kuznets curve is a hypothetical curve that shows the relationship between income inequality and economic growth. It suggests that as an economy develops, income inequality initially increases, but eventually decreases as the economy reaches a certain level of maturity. The curve is inverted U-shaped, with inequality increasing in the early stages of development and decreasing in the later stages.



D) The Laffer curve, on the other hand, shows the relationship between tax rates and government revenue. It suggests that as tax rates increase, revenue initially increases, but eventually decreases as high tax rates discourage economic activity.

21-A:

Cyclical unemployment (A) is associated with a deficiency in aggregate demand, which occurs during economic downturns or recessions. This type of unemployment is caused by a lack of demand for labor, resulting in layoffs and job losses.

B) Frictional unemployment refers to the temporary unemployment experienced by workers transitioning between jobs.

C) Structural unemployment is caused as a result of changes in the structure of the economy, that is when the pattern of demand and supply changes.

D) Voluntary unemployment refers to situations where workers choose not to work at the current market wage rate.

22-C:

The shift in demand from labor-intensive to high-technology production methods has led to unemployment among workers in the labor-intensive sector. Subsidizing products produced by labor-intensive methods can help induce demand for these products, encouraging firms to hire more workers and reducing unemployment. **This explains why key C will be correct.**

A) Reducing income tax rates may **cure “demand deficient unemployment”** and not specifically address the unemployment issue in the labor-intensive sector.

B) Increasing interest rates on borrowing for investment could actually discourage firms from investing in new technologies and hiring workers. This contractionary monetary policy will bring about an increase in demand deficient unemployment in the economy.

D) Subsidizing exports produced by high-technology industries could exacerbate the problem by further promoting the shift away from labor-intensive methods.

23-D:

The correct answer is D) indirect taxation.

In a **free-market** economy, the **government's role is generally limited**. Indirect taxation, such as sales taxes or value-added taxes (VAT), is a government intervention that would not exist in a purely free-market economy.

A) Autonomous investment refers to investment decisions made by firms based on their own expectations (**which are independent on national income**), which would exist in a free-market economy.

B) Household saving is the decision of households to set aside a portion of their income, which is a natural aspect of a free-market economy.

C) Import spending is the act of buying goods and services from foreign countries, which occurs **in an open economy**.

Key takeaway: An open economy is open to international trade.

24- C:

An economy with a **positive output gap**, also known as an inflationary gap, is producing above its potential GDP. This means that there is excess of **Aggregate demand** in the economy **relative to Aggregate Supply**, leading to upward pressure on prices and, consequently, **demand-pull** inflation. (key C)

A) A positive output gap means that actual GDP is higher than potential GDP, so this option is incorrect.

B) High unemployment is more likely to be associated with a negative output gap (recessionary gap), not a positive one.

D) A positive output gap may lead to higher labor costs due to increased demand for labor.

25-D:

-> When the interest rate on new bonds increases, the market price of old bonds decreases, because the old bonds are less attractive compared to the new bonds with a higher interest rate.

Conversely, when the interest rate on new bonds decreases, the market price of old bonds increases, because the old bonds are more attractive compared to the new bonds with a lower interest rate.

So, the correct explanation is:

- Interest rate $\uparrow$ $\rightarrow$ Market price of old bonds $\downarrow$ (old bonds less attractive)
- Interest rate $\downarrow$ $\rightarrow$ Market price of old bonds $\uparrow$ (old bonds more attractive)

Why does this happen?

The interest rate (or yield) of a bond determines the return on investment for the bondholder. When interest rates change, the expected return on investment also changes, affecting the bond's price.

A) The transaction and precautionary motives are indeed interest inelastic, meaning they are weakly influenced by interest rates and more directly related to the level of income. People hold money for transactions and precautionary purposes regardless of interest rates.

B) According to Keynes, the money demand (liquidity preference) is shallow and highly elastic at low levels of interest. In this case, monetary policy (e.g., increasing the money supply) has little effect on stimulating the economy.

C) The demand for money by households is not necessarily stable. It can be affected by various factors, including changes in income, expenses, uncertainty, and interest rates.

26-B:

MPC (Marginal Propensity to Consume) is the change in consumption divided by the change in income. Using the formula:

$$\text{MPC} = \text{change in consumption} / \text{change in income}$$

We can calculate the MPC for each interval:

Income	Consumption	ΔC	ΔI	MPC
100	140	-	-	-
200	220	80	100	0.8
300	300	80	100	0.8
400	380	80	100	0.8
500	460	80	100	0.8

As you can see, the MPC is constant at 0.8.

APC (Average Propensity to Consume) is consumption divided by income. Using the formula:

$$\text{APC} = \text{consumption} / \text{income}$$

We can calculate the APC for each interval:

Income	Consumption	APC
100	140	1.4
200	220	1.1
300	300	1.0
400	380	0.95
500	460	0.92

As you can see, the APC is falling as income rises.

This explains why key B will be correct.

27-B:

- Foreign Direct Investment is not considered foreign aid, as it's a private investment made by a company in a foreign country.
- Official Development Assistance is indeed considered foreign aid, as it's a type of assistance provided by governments or official agencies to support development in other countries.
- Portfolio Investment is not considered foreign aid, as it's a type of investment in foreign financial assets, such as stocks or bonds, made by private individuals or companies.

28-C:

When interest rates increase, **costs of borrowing increase** and it becomes more expensive for businesses to borrow money to invest in new projects or expand existing ones. This leads to a decrease in investment spending.

- An increase in business confidence (A) would lead to an increase in investment spending, as businesses are more likely to invest when they are optimistic about the future.
- An increase in company profits (B) would also lead to an increase in investment spending, as businesses are more likely to re-invest their **retained profits**.
- An increase in national income (D) would lead to an increase in investment spending, as a growing economy provides more opportunities for investment. **Link with accelerator theory.**

29-D:

The Laffer curve is a theoretical representation of the relationship between tax rates and government revenue. It suggests that as tax rates increase, revenue initially increases, but eventually decreases as high tax rates discourage economic activity. This is why only key D can be correct.

A – There's no such curve as income elasticity curve.

B – J curve shows the impact of devaluation on Balance of trade position over time.

C -The Lorenz Curve is a graphical representation of income inequality in a population. It's a cumulative distribution curve that shows the proportion of total income earned by different percentages of the population.

30-D:

The marginal effect is basically the impact of a change in value on one variable when the other variable's value is changed.

In this question, if you look below , you will realise that the inflation rate does have a marginal effect on the rate of unemployment. What this means is that when the inflation rate observes a change of 0.3, the rate of unemployment observes a change of 0.1.

This ratio can be easily observed till the table end whenever there was a change of 0.3 in the inflation rate, the unemployment rate changed by 0.1, therefore, this ratio remained constant.

Hence, it is correct to conclude that inflation rate has a marginal effect on the rate of unemployment that they follow a fixed ratio pattern

month	April	May	June	July	Aug	Sept
rate of inflation (%)	2.2	1.9	1.6	1.7	1.9	2.2
rate of unemployment (%)	3.9	3.8	3.9	3.9	3.8	3.7

Only key D can be correct.

